APPENDIX E2
SITE SAFETY AND ENVIRONMENTAL CYCLE
[PS Clause 1.13A]

1. DAILY CYCLE

1.1 PES Meeting

1.1.1 The *Contractor* shall arrange and hold PES meetings each about 10 to 15 minutes for all the persons employed on the *works*.

1.1.2 The PES meeting shall be led by the Site Agent or a senior staff of site management of the *Contractor*, who has attended the training course on SSEC or the Safe and Environmental Working Cycle of the Occupational Safety & Health Council (OSHC) or Construction Industry Training Authority (CICTA), or other relevant training courses notified by the *Supervisor*.

1.1.3 The first few minutes of PES meeting shall start with a physical exercise set by the *Contractor*. After that, the leader of the PES meeting shall address the attendees on the prevailing safety and health matters related to the Site, such as common hazards and control measures, general fire and safety precautions, specific safety concerns, general defects and irregularities observed in inspections, accidents or near misses etc. Besides, the *Contractor* shall also make use of the PES meetings to announce common safety matters in execution and co-ordination of the *works* on the Site among sub-contractors and workers, or presenting awards to workers and/or sub-contractors in recognition of their good safety performance.

1.1.4 The *Contractor* shall maintain a brief record of the run-down programme and a register of the persons attended for each PES meeting. The record and the register shall be kept in the *Contractor*'s site office for ready inspection by the *Supervisor* or his/her staff upon request. The *Contractor* can use separate sheets for registering attendees in order to save time for taking attendance in the meeting. The *Supervisor* and/or his/her staff shall attend the PES meetings regularly to ensure their quality for certifying payment.

1.2 HIA Meeting

1.2.1 The *Contractor* shall arrange and hold HIA meeting of about 10 minutes for the attendees immediately after the PES meeting.

1.2.2 To strengthen the communication on the Site, the *Contractor* shall arrange persons on the Site to take turn to lead the HIA meeting. Such person can be either a foreman, ganger, Safety Officer, Safety Supervisor or Safety Representative who has attended, in addition to the SSEC course as mentioned in sub-clause 1.1.2 above, the presentation skill course such as the Occupational Safety and Health Trainer Course of OSHC or the Safety Training Techniques of CICTA or other relevant courses notified by the *Supervisor*. The *Contractor* shall ensure that sufficient persons on the Site have received the training to lead the HIA meetings as soon as the Contract commences.

1.2.3 The leader of each HIA meeting shall prepare the training materials before conduction, which shall include, but is not limited to, hazards and control measures specific to the works or trades, special safety concerns, assurance of safety requirements and measures, reprimand of repeated irregularities and malpractice etc. Besides, the leaders of the HIA meetings shall also encourage workers to give their views in the HIA meetings. Where necessary, Site Agent, Safety Officer and/or Safety Supervisors who are more experienced in provision of training shall provide guidance and assistance to the leaders of HIA meeting before conducting the meeting. The training materials prepared for and the discussion during the HIA meeting shall be recorded in a HIA table, a sample of which is shown in Annex A. The HIA Table shall be kept in the *Contractor*'s site office for ready inspection by the *Supervisor* or his/her staff upon request.

1.2.4 Subject to the agreement of the *Supervisor*, the *Contractor* can alternatively arrange the HIA meetings to be held in small groups according to the trades, work teams or works areas set out by the *Contractor* for the Site. If so, the *Contractor* shall ensure that the assigned persons are competent to lead the HIA meetings, whose names and curriculum vitae shall be submitted to the *Supervisor* for approval. The *Supervisor* or his/her staff shall attend the HIA meetings regularly to ensure their quality for certifying payment. The *Supervisor* shall not certify payment for the number of persons who have attended the HIA meeting in a group if the *Supervisor* is dissatisfied with the content and/or the arrangement of the HIA meeting for that group.

1.2.5 The *Contractor* can use the register for the PES meeting for taking attendance of the HIA meeting if the attendees have not split up into small groups. Otherwise, the *Contractor* shall use separate sheets for registering attendees of individual groups of the HIA meetings, and attach to the register of the PES meeting held on the same day for ready inspection by the *Supervisor* for his/her staff upon request. Furthermore, the *Contractor* shall complete one HIA table for each HIA meeting held for individual groups pursuant to sub-clause 1.2.4 above.

1.3 Pre-work Safety Checks

1.3.1 The *Contractor* shall arrange and hold Pre-work Safety Checks for the attendees immediately after the HIA meeting. The Pre-work Safety Checks shall be carried out by foremen, gangers, Safety Supervisors or Safety Representatives of the attendees according to the trades, work teams or works areas set out by the *Contractor* for the Site. The Pre-work Safety Checks shall include the checking of personal protective equipment worn by attendees before they start working on that day such as safety helmet, reflective vest, ear protectors, eye protectors, safety harness, safety footwear etc. The *Supervisor* or his/her staff shall attend the Pre-work Safety Checks regularly to ensure the proper checking by the *Contractor* for certifying payment.

1.3.2 The *Contractor* shall assign persons who are competent with the relevant knowledge, experience and training to check and ascertain the safety conditions of facilities, machinery, plant and equipment and materials before commencing work on that day. The *Contractor* shall propose a list of facilities, machinery, plant and equipment to be checked and develop relevant checklists for such checking for the approval of the *Supervisor*. The assigned persons shall use the checklists for Pre-work Safety Checks of facilities, machinery, plant and equipment, and the completed checklists shall be kept at the *Contractor*'s site office for ready inspection by the *Supervisor* or his/her staff upon request.

1.4 Safety Inspection by the Site Agent or his Representative

1.4.1 The *Contractor* shall arrange the Site Agent or his/her representative to carry out safety inspection of the Site daily, particularly for those areas identified for improvements in weekly safety co-ordination meetings, Weekly Safety Walks, Site Safety Management Committee (SSMC) meetings or Site Safety Committee meetings pursuant to sub-clause 2.1, 2.2, 3.1 and 3.2 respectively. The Site Agent or his/her representative shall check and ensure that the safety instructions given in PES meetings or HIA meetings have been observed and carried out.

1.4.2 Any unsafe act or unsafe conditions observed during inspections shall be recorded in a diary maintained by the Site Agent, who shall promptly communicate the irregularity to the respective party concerned for follow-up actions. The Site Agent shall check and ensure that the unsafe acts or unsafe conditions are rectified promptly, and the date is duly recorded in the diary after completion. The safety diary shall be made available for inspection by the *Supervisor* and copying thereof upon request.

1.5 Guidance and Supervision during Work

1.5.1 The *Contractor* shall assign sufficient supervisory staff to be responsible for the safety and health of workers on the Site. The names of the assigned supervisory staff shall be shown in the site

safety and Environmental organization chart posted up in the safety bulletin board pursuant to sub-clause 4.2 below. The assigned supervisory staff shall provide guidance and supervision for the workers under his/here control, and rectify any irregularities, unsafe acts or unsafe conditions for the works on the Site. Guidance and supervision provided shall also include the implementation of safety instructions given in PES or HIA meetings.

1.6 Safety Co-ordination Meeting

1.6.1 The *Contractor* shall arrange and hold safety co-ordination meeting each day to coordinate safety and health work to be carried out on the Site on the following day. The meeting shall be chaired by the Site Agent or a senior staff of site management of the *Contractor*, and attended by assigned supervisory staff pursuant to sub-clause 1.5 above where necessary. The meeting shall be used to discuss the findings in safety inspections and/or the matters to be announced in the next PES or HIA meeting. It can also be used for discussion and co-ordination of site safety matters, such as sequent of works, usage times for shared machinery and equipment and works areas phasing of works at various interfaces, delivery and storage of materials and equipment to the Site etc.

1.7 Daily Cleaning and Tidying up of the Site

1.7.1 Detailed requirements are specified in PS Clause No. 1.32A.

1.8 Checking of the Site after Each Day's Work

1.8.1 The *Contractor* shall assign designated person to check the safety of the Site after each day's work including, but is not limited to, the following:

(i) all flames and heat sources have been extinguished (particularly for welding and hot-work operations);

(ii) keys have been removed from construction machinery and plant and kept in a safe place;

(iii) all construction plant are parked properly on level and stable ground;

(iv) all machinery and power sources have been turned off;

(v) all openings are properly covered and all edges are provided with fall protection measures;

(vi) all nailed timber, planks and/or sheets are piled and put aside away from main accesses;

(vii) excavated surfaces on soil slopes are provided with temporary protection;

(viii) all temporary works are properly maintained;

(ix) signing, lighting and guarding are provided in accordance with the approved temporary traffic management scheme;

(x) hoarding and/or covered walkways along the periphery of the Site are maintained in proper condition; and

(xi) the Site has been fenced and guarded against unauthorized entry.

The designated person shall, after completion of checking, notify the Site Agent any unsafe conditions or imminent danger that require immediate follow-up actions. The designated person shall also draw the attention of the Site Agent about the minor irregularities to arrange rectification on the following day.

2. WEEKLY CYCLE

2.1 Weekly Safety Walk

2.1.1 Detailed requirements are specified in Clause 11 of Appendix E1.

2.2 Weekly Safety Co-ordination Meeting

2.2.1 The *Contractor* shall arrange the Site Agent or a senior staff of the site management of the *Contractor* together with the Safety Manager, the Safety Officer, Safety Supervisor and/or supervisory staff of sub-contractors to attend the weekly safety co-ordination meeting chaired by the *Supervisor*. The meeting shall discuss or coordinate safety and health matters including, but is not limited to, safety performance, housekeeping and tidiness of the Site, together with the specific areas of concern, defects and deficiencies observed in Weekly Safety Walks, accidents and near misses occurred on the Site, etc. A brief notes of the meetings shall be prepared by the *Contractor* and endorsed by the *Supervisor* after the Meeting.

2.3 Weekly Overall Cleaning and Tidying Up of the Site

2.3.1 Details requirements are specified in PS Clause No. 1.32A.

3. MONTHLY CYCLE

3.1 Site Safety Management Committee (SSMC)

3.1.1 Detailed requirements are specified in Clause 10 of Appendix E1.

3.2 Site Safety Committee Meetings (SSC)

3.2.1 Detailed requirements are specified in Clause 9 of Appendix E1.

4. SAFETY BULLETIN BOARD

4.1 The *Contractor* shall provide a safety bulletin board at the location where Pre-work Activities pursuant to sub-clauses 1.1 to 1.3 shall be held. Subject to the approval of the *Supervisor*, additional safety bulletin board can be provided where Pre-work Activities have to be held at more than one location of the Site.

4.2 The safety bulletin board shall be made of recycled materials with a size approximately 6m x 1.5m which can be varied to suit the site condition subject to the approval by the *Supervisor*. The board shall contain, but is not limited to, the following information:

(i) the design by the *Contractor* on promotion of Site Safety and Environmental Cycle;

(ii) a chart with names and contact telephone numbers showing the site safety organizational structure from senior site management down to sub-contractors, gangers, foremen, Safety Officer, Safety Supervisors and Safety Representatives according to works trades, work gangs or works areas, together with the emergency teams, first aid personnel etc.

(iii) accident statistics with breakdown to sub-contractors; and

(iv) a figure showing a worker wearing all the personal protective equipment plus a mirror to cover the full view of the person when standing in front.

The safety bulletin board may contain other safety information related to the Site such as Site plan, safety policies, in-house safety rules and regulations, slogans, colour coding systems for lifting gears, important safety and health issues etc.

4.3 The *Contractor* shall propose the location for the erection of the safety bulletin board and the hard-paved area pursuant to sub-clause 5 where Pre-work Activities will be held for the approval of the *Supervisor*.

5. HARD-PAVED AREA

5.1 The *Contractor* shall arrange and provide a hard-paved area on the Site for holding Pre-work Activities pursuant to sub-clauses 1.1 to 1.3 above under the provision of *Contractor*'s Site Accommodation in the Preliminaries. The hard-paved area shall be sized based on a rate of 1.5m² per person multiplied by the maximum number of persons to be arranged for one session of Pre-work Activities. The hard-paved area shall be designed to be able to sustain the load at that area throughout the Contract. Details of construction of the hard-paved area shall be submitted to the *Supervisor* for approval.

5.2 Subject to the agreement of the *Supervisor*, the *Contractor* can arrange Pre-work Activities to be held at more than one location, and the hard-paved areas at each venue shall be sized separately based on the specified requirement given in sub-clause 5.1 above. The hard-paved area can be an open area in front of the site office, or an area inside building structure under construction, or at the main works areas in case of roadworks or mains laying contracts, or near shaft opening for a tunnelling contract, whichever is appropriate. The *Contractor* shall submit the proposed location and its area for holding Pre-work Activities for the approval of the *Supervisor* within 14 days from the *starting date* on the Site.

5.3 If the proposed location for holding of Pre-work Activities is outdoors, consideration shall be given to provide a cover for the hard-paved area unless otherwise approved by the *Supervisor* that the provision is not necessary. Details of construction of the cover shall be submitted to the *Supervisor* for approval.

5.4 The *Contractor* shall maintain the hard-paved area and the cover throughout the Contract, and shall demolish and reinstate the area to the satisfaction of the *Supervisor* prior to completion of the *works* unless otherwise agreed by the *Supervisor*.

Record of Hazard Identification Activity (HIA) Meeting

Contract No. : Name of *Contractor* :

Contract Title :

Annex A

Team Reference *(if appropriate)*:	
Date:	Time:
Potential Hazards / Irregularities / Unsafe Acts or Unsafe Conditions	Safety Precautionary Measures / Rectification Actions / Improvement Actions
Name of Leader:	
Signature:	Position:

(Please complete one sheet for one HIA Meeting)